

Bootstrap Utilities Assignment

Question 1: Flipkart Product Card with Bootstrap Spacing Utilities

Create an HTML page that uses Bootstrap utility classes to add 3 different levels of spacing (margin and padding) around a card displaying a Flipkart-style product image and name.

product-spacing.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Flipkart Card - Spacing Utilities</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <div class="container">

    <!-- Level 1: small spacing (m-2 / p-2) -->
    <div class="card m-2 p-2" style="width: 16rem;">
      
      <div class="card-body">
        <p class="card-title mb-0">Running Shoes</p>
      </div>
    </div>

    <!-- Level 2: medium spacing (m-4 / p-4) -->
    <div class="card m-4 p-4" style="width: 16rem;">
      
      <div class="card-body">
        <p class="card-title mb-0">Smart Watch</p>
      </div>
    </div>

    <!-- Level 3: large spacing (m-5 / p-5) -->
    <div class="card m-5 p-5" style="width: 16rem;">
      
      <div class="card-body">
        <p class="card-title mb-0">Laptop Backpack</p>
      </div>
    </div>

  </div>

</body>
</html>
```

Explanation: Bootstrap's spacing utilities follow the pattern `{property}{sides}-{size}`, where `m` is margin and `p` is padding, with size values from 0 to 5. The three cards use increasing spacing levels — `m-2 p-2` (small), `m-4 p-4` (medium), and `m-5 p-5` (large) — so each card visibly has more breathing room around and inside it than the last, demonstrating three different spacing levels without writing any custom CSS.

Question 2: BookMyShow-style Movie Review Card (Typography Utilities)

Build a movie review card (like BookMyShow) using Bootstrap typography utilities to style the movie title, rating, and review text with different font weights and sizes.

movie-review-card.html

```
<!DOCTYPE html>
<html>
<head>
  <title>BookMyShow Style Review Card</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <div class="container mt-4">
    <div class="card p-3" style="max-width: 400px;">

      <h3 class="fw-bold fs-3">Pathaan</h3>

      <p class="fw-semibold fs-5 text-warning">★ 4.3 / 5</p>

      <p class="fw-normal fs-6 text-secondary">
        A high-octane action thriller with stunning visuals and a
        gripping storyline. Shah Rukh Khan delivers a powerful comeback
        performance. Definitely worth watching on the big screen.
      </p>

    </div>
  </div>

</body>
</html>
```

Explanation: The movie title uses `fw-bold fs-3` for a large, bold heading. The rating uses `fw-semibold fs-5` with `text-warning` (a gold/yellow tone) to draw attention as a mid-sized, semi-bold highlight. The review text uses `fw-normal fs-6` with `text-secondary` (muted gray), making it the smallest and lightest-weight text on the card — creating a clear visual hierarchy: title > rating > review, purely through Bootstrap's `fw-*` (font-weight) and `fs-*` (font-size) utility classes.

Question 3: Zomato-style Restaurant List (Color Utilities)

Design a Zomato-style restaurant list where each restaurant card uses at least three different Bootstrap color utility classes to highlight ratings, offers, and open/closed status.

restaurant-list.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Zomato Style Restaurant List</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <div class="container mt-4">

    <div class="card mb-3 p-3">
      <h5>Spice Villa</h5>
      <p>
        <span class="badge bg-success">★ 4.3</span>      <!-- rating: green -->
        <span class="badge bg-warning text-dark">50% OFF</span>  <!-- offer: yellow -->
        <span class="badge bg-danger">Closed</span>      <!-- status: red -->
      </p>
    </div>
  </div>

</body>
</html>
```

```

    </p>
  </div>

  <div class="card mb-3 p-3">
    <h5>Urban Tandoor</h5>
    <p>
      <span class="badge bg-success">★ 4.6</span>
      <span class="badge bg-warning text-dark">Flat ₹100 OFF</span>
      <span class="badge bg-primary">Open Now</span>      <!-- status: blue/open -->
    </p>
  </div>

</div>

</body>
</html>

```

Explanation: Each restaurant card uses Bootstrap's `badge bg-*` color utility classes for three different pieces of information: `bg-success` (green) highlights the rating as a positive signal, `bg-warning` (yellow) draws attention to the current offer/discount, and `bg-danger` or `bg-primary` is used to distinguish "Closed" (red, a warning) from "Open Now" (blue, a neutral/positive status) — giving each badge a distinct color meaning at a glance, similar to how Zomato visually differentiates this information.

Question 4: Custom 'Book Ticket' Button (Overriding Bootstrap Primary Color)

Override the Bootstrap primary button color in your own CSS file to use a custom shade (not the default blue), and apply it to a 'Book Ticket' button on your page.

index.html

```

<!DOCTYPE html>
<html>
<head>
  <title>Custom Book Ticket Button</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
  <!-- custom.css is linked AFTER Bootstrap so it can override .btn-primary -->
  <link href="custom.css" rel="stylesheet">
</head>
<body>

  <div class="container mt-4 text-center">
    <button class="btn btn-primary btn-lg">Book Ticket</button>
  </div>

</body>
</html>

```

custom.css

```

/* Overrides Bootstrap's default blue .btn-primary with a custom red-orange shade */
.btn-primary {
  background-color: #e64a19;
  border-color: #e64a19;
}

.btn-primary:hover {
  background-color: #bf360c;
  border-color: #bf360c;
}

```

Explanation: Because `custom.css` is linked in the HTML *after* the Bootstrap stylesheet, its `.btn-primary` rule has higher precedence in the cascade (same specificity, but declared later wins) and overrides Bootstrap's default blue with a custom red-orange color (`#e64a19`). A matching `:hover` rule is also overridden so the darker hover shade stays consistent with the new custom color instead of reverting to Bootstrap's default blue hover.

Question 5: Light/Dark Theme Switcher (Bootstrap Background & Text Classes)

Add a theme switcher button to your page that toggles between a light and dark background using Bootstrap classes, and updates the text color for readability.

theme-switcher.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Theme Switcher</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <div id="mainContainer" class="bg-light text-dark min-vh-100 p-4">

    <h2>Welcome to the App</h2>
    <p>This page supports both light and dark themes.</p>

    <button id="themeToggle" class="btn btn-outline-secondary">
      Toggle Dark Mode
    </button>

  </div>

  <script>
    const container = document.getElementById('mainContainer');
    const toggleBtn = document.getElementById('themeToggle');

    toggleBtn.addEventListener('click', () => {
      container.classList.toggle('bg-light');
      container.classList.toggle('bg-dark');
      container.classList.toggle('text-dark');
      container.classList.toggle('text-light');
    });
  </script>

</body>
</html>
```

Explanation: The main container starts with Bootstrap's `bg-light` and `text-dark` classes for the default light theme. Clicking the toggle button runs a small JavaScript function that uses `classList.toggle()` on all four classes at once: `bg-light/bg-dark` switch the background between white/light-gray and near-black, while `text-dark/text-light` switch the text color between dark and white so it always stays readable against whichever background is currently active. Clicking the button again reverses all four classes back to the light theme.